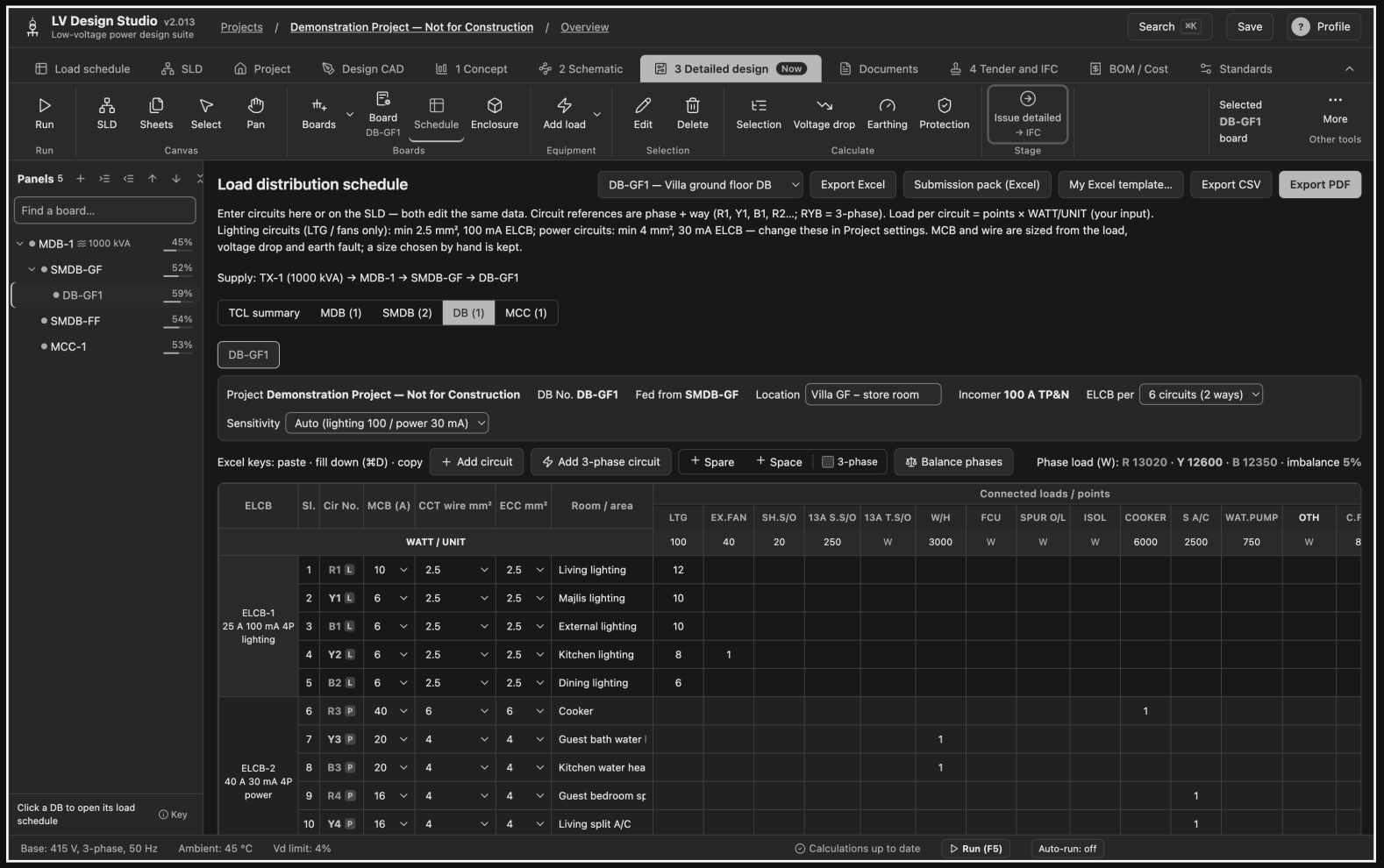
Load Schedule

Visual user manual - enter circuits, balance phases, review demand, and export



01 Open the correct panel

02 Complete the DB schedule

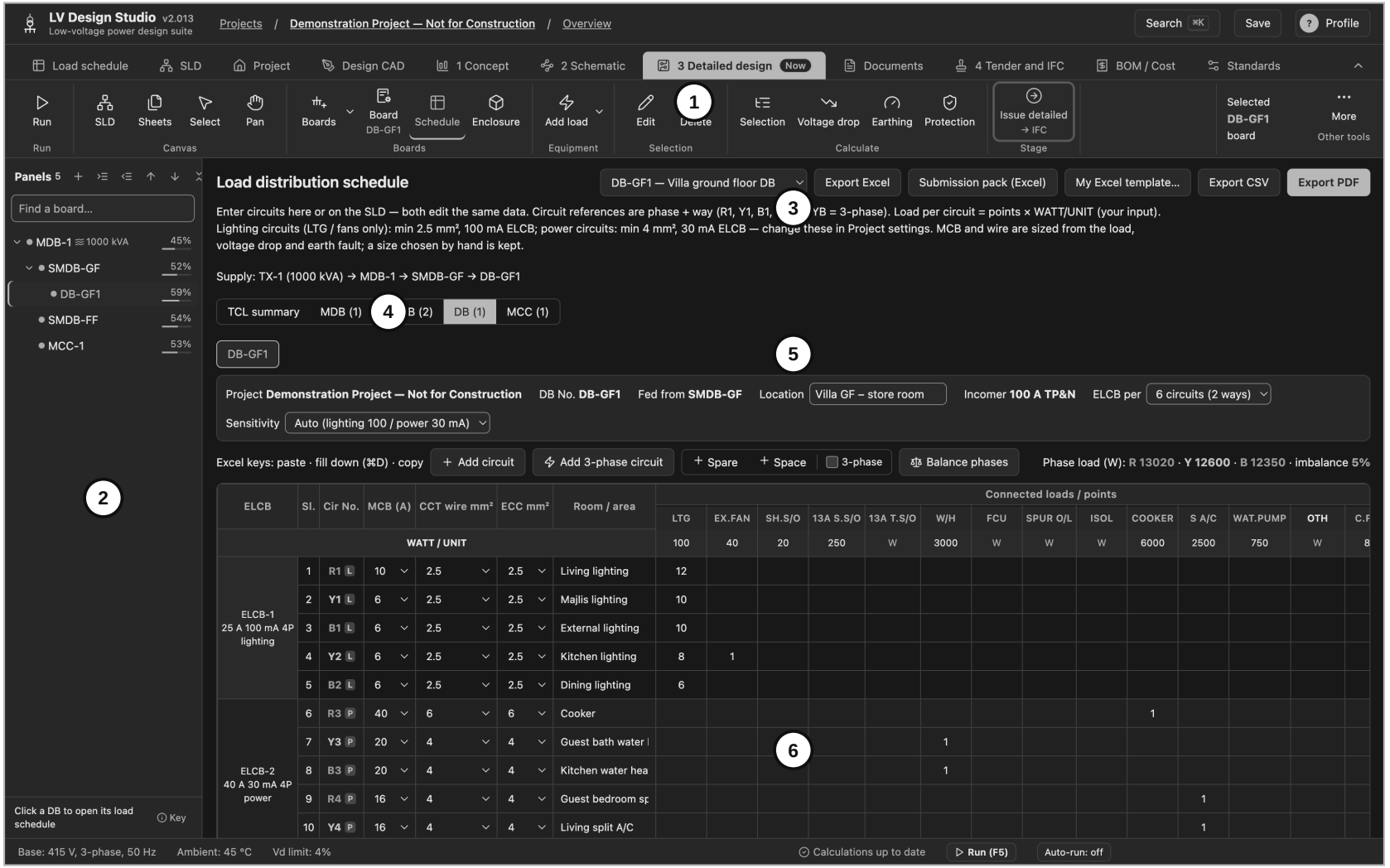
03 Review MD and phase balance

04 Export submission forms

START HERE

Know the screen before entering data

The numbered markers identify the controls used most often.



1. Schedule button in Detailed design
2. Panel tree - choose the board
3. Panel selector and current form
4. Form groups and panel tabs
5. Excel, PDF, and submission exports
6. Editable schedule or MD grid

STEP 1

Open the panel and form you intend to edit

Do not type a downstream panel load into its parent form. Edit the source schedule instead.

1 Select a board

Click a board in the panel tree, use the board selector, or right-click the board and choose Load schedule.

2 Choose the form group

Use TCL summary, MDB, SMDB, DB, MCC, or Busbar riser. An exclamation mark means the form is not filled yet.

3 Choose Load schedule or MD

DB final circuits belong in Load schedule. Panel incomers and outgoing feeders belong in Connected load & MD.



Quick decision

- Final lighting, socket, A/C, water-heater, and equipment circuits: use the DB Load schedule.
- MDB/SMDB/MCC incomer and outgoing feeders: use Connected load & MD.
- Transformer roll-up: use TCL summary. High-rise tap-offs: use Busbar riser.
- If a feeder supplies another panel, edit that panel or its DB schedules; the totals roll up automatically.

STEPS 2-5

Enter points, circuits, protection, and length

The grid below is an interface illustration using the same fields as the app.

DB-GF1 - Load distribution schedule

ELCB per: 6 circuits Sensitivity: Auto

WATT / UNI (1)	LTG 100	FAN 40	13A 250	WH 3000	SAC (2)	PUMP 750

(3) ELCB	Sl.	Circuit	MCB	Wire	Room / area	Connected loads / points	R	Y	B
ELCB-1	1	R1	10	2.5	Living lighting	12 LTG	1200		
	2	Y1	10	2.5	(4) Majlis lighting	10 LTG		1000	
	3	B1	16	4	Split A/C	1 SAC			2500
ELCB-2	4	R2	20	4	Water heater	1 WH	3000		
	5	Y2	16	4	Sockets	6 x 13A		1500 (5)	

1. Enter WATT / UNIT for every point type used
2. Use a library item when a standard load is available
3. Add circuit or Add 3-phase circuit
4. Enter room and point quantities
5. Enter length and review Check

STEPS 6-8

Finish the circuit details and review the board

Double-click a circuit row for the detailed editor. Use Balance phases only after the circuit list is ready.

Edit circuit: R1

DB-GF1 - phase R - changes apply on Save

Room / area
Living room

Length (m) - calc only
22

Lighting points
12

Socket points
0

MCB (A)
10 - automatic

CCT wire mm2
2.5 - automatic

- Use load items when one circuit contains mixed wattages.
- Clear Sized by hand to return MCB, wire, and ECC to automatic sizing.
- Add Spare for a fitted breaker; add Space for an empty future way.

Board review

1 Balance phases

Spreads circuits across R/Y/B, groups lighting before power, renumbers circuits, and moves spares to free slots.

2 Read phase totals

Review R, Y, and B watts and the imbalance percentage. Values above 20% are highlighted for checking.

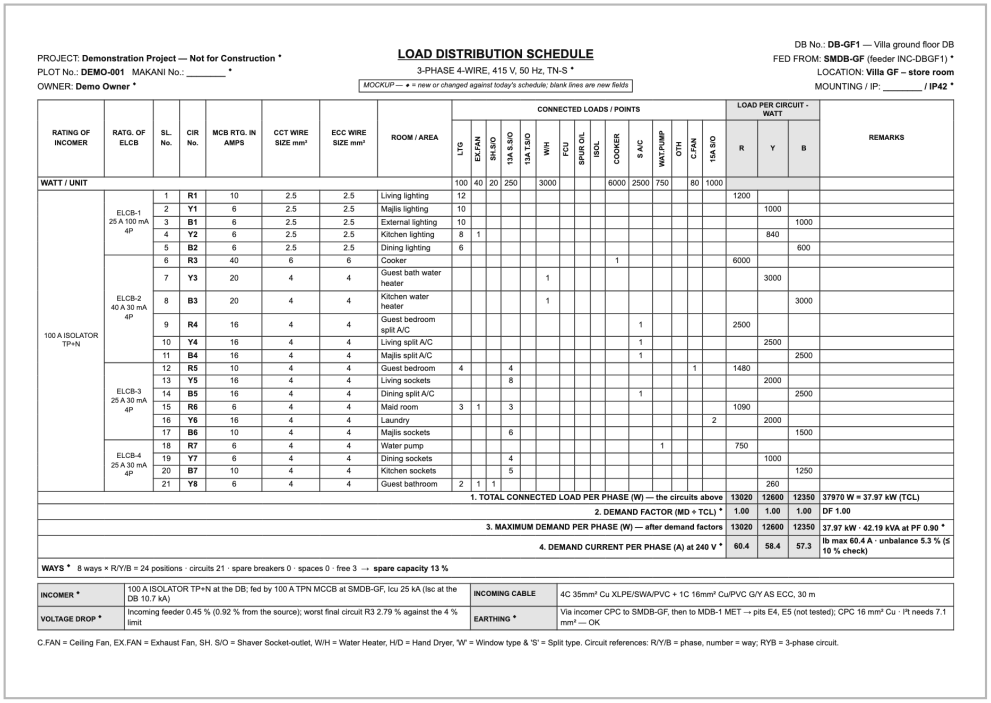
3 Resolve checks

Fix missing watts, wire below minimum, mixed ELCB groups, invalid lengths, and other rejected edits.

FINISH

Review the output, then export the right file

The example below is an illustrative schedule-output mockup, not a live app capture.



PROJECT: **Demonstration Project — Not for Construction ***
PLOT No.: **DEMO-001** MAKANI No.: _____ *
OWNER: **Demo Owner ***

LOAD DISTRIBUTION SCHEDULE
3-PHASE 4-WIRE, 415 V, 50 Hz, TN-S *
MOCKUP — ★ = new or changed against today's schedule, blank lines are new fields

DB No.: **DB-GF1** — Villa ground floor DB
FED FROM: **SMDB-GF** (feeder INC-DBGF1) *
LOCATION: **Villa GF – store room**
MOUNTING / IP: _____ / **IP42** *

RATING OF INCOMER	RATG. OF ELCB	SL. No.	CIR No.	MCB RTG. IN AMPS	CCT WIRE SIZE mm²	ECC WIRE SIZE mm²	ROOM / AREA	CONNECTED LOADS / POINTS														LOAD PER CIRCUIT - WATT			REMARKS
								LTG	EX.FAN	SH.S/O	13A S.S/O	13A T.S/O	WHH	FCU	SPUR O/L	ISOL	COOKER	S A/C	WAT.PUMP	OTH	C.FAN	15A S/O	R	Y	B	
WATT / UNIT								100	40	20	250		3000				6000	2500	750		80	1000				
100 A ISOLATOR TP+N	ELCB-1 25 A 100 mA 4P	1	**R1**	10	2.5	2.5	Living lighting	12															1200			
		2	**Y1**	6	2.5	2.5	Majlis lighting	10																1000		
		3	**B1**	6	2.5	2.5	External lighting	10																	1000	
		4	**Y2**	6	2.5	2.5	Kitchen lighting	8	1															840		
		5	**B2**	6	2.5	2.5	Dining lighting	6																	600	
	ELCB-2 40 A 30 mA 4P	6	**R3**	40	6	6	Cooker										1						6000			
		7	**Y3**	20	4	4	Guest bath water heater						1											3000		
		8	**B3**	20	4	4	Kitchen water heater						1												3000	
		9	**R4**	16	4	4	Guest bedroom split A/C											1					2500			
		10	**Y4**	16	4	4	Living split A/C											1						2500		
		11	**B4**	16	4	4	Majlis split A/C											1							2500	
	ELCB-3 25 A 30 mA 4P	12	**R5**	10	4	4	Guest bedroom	4			4										1		1480			
		13	**Y5**	16	4	4	Living sockets				8													2000		
		14	**B5**	16	4	4	Dining split A/C											1							2500	
		15	**R6**	6	4	4	Maid room	3	1		3												1090			
		16	**Y6**	16	4	4	Laundry															2		2000		
		17	**B6**	10	4	4	Majlis sockets				6														1500	
	ELCB-4 25 A 30 mA 4P	18	**R7**	6	4	4	Water pump												1				750			
		19	**Y7**	6	4	4	Dining sockets				4													1000		
		20	**B7**	10	4	4	Kitchen sockets				5														1250	
		21	**Y8**	6	4	4	Guest bathroom	2	1	1														260		
							1. TOTAL CONNECTED LOAD PER PHASE (W) — the circuits above																	**13020**	**12600**	**12350**	**37970 W = 37.97 kW (TCL)**
							2. DEMAND FACTOR (MD ÷ TCL) *																	**1.00**	**1.00**	**1.00**	**DF 1.00**
							3. MAXIMUM DEMAND PER PHASE (W) — after demand factors																	**13020**	**12600**	**12350**	**37.97 kW · 42.19 kVA at PF 0.90 ***
							4. DEMAND CURRENT PER PHASE (A) at 240 V *																	**60.4**	**58.4**	**57.3**	**Ib max 60.4 A · unbalance 5.3 % (≤ 10 % check)**

WAYS * 8 ways × R/Y/B = 24 positions · circuits 21 · spare breakers 0 · spaces 0 · free 3 → **spare capacity 13 %**

INCOMER *	100 A ISOLATOR TP+N at the DB; fed by 100 A TPN MCCB at SMDB-GF, Icu 25 kA (Isc at the DB 10.7 kA)	**INCOMING CABLE**	4C 35mm² Cu XLPE/SWA/PVC + 1C 16mm² Cu/PVC G/Y AS ECC, 30 m
VOLTAGE DROP *	Incoming feeder 0.45 % (0.92 % from the source); worst final circuit R3 2.79 % against the 4 % limit	**EARTHING ***	Via incomer CPC to SMDB-GF, then to MDB-1 MET → pits E4, E5 (not tested); CPC 16 mm² Cu · I²t needs 7.1 mm² — OK

C.FAN = Ceiling Fan, EX.FAN = Exhaust Fan, SH. S/O = Shaver Socket-outlet, W/H = Water Heater, H/D = Hand Dryer, 'W' = Window type & 'S' = Split type. Circuit references: R/Y/B = phase, number = way; RYB = 3-phase circuit.

Illustrative load distribution schedule output

Before export

- No missing WATT / UNIT warnings
- No incomplete-panel exclamation marks
- R/Y/B balance reviewed
- Circuit Check column reviewed
- Project and submission details complete
- Calculations run with F5
- Project saved

Choose an export

- Export Excel - current form
- Export PDF - current form
- Export CSV - current DB grid
- Submission pack (Excel) - all forms